Law of Admissibility

Structural Invariants in Recursive Dynamics ($R = 4$)

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Abstract

The Law of Admissibility locates the step at which a recursive system of memory depth k completes four full turns of its slowest damped mode. For the unweighted k-nacci families, the characteristic roots fix the period P of that mode, and the period of the slowest mode satisfies $k-1 < P \leq k$ for every $k \geq 2$, with equality only for Fibonacci. It follows that $3k < 4P \leq 4k$, so $n = 4k$ is the first window boundary after four full turns. This step is the admissibility boundary, and the $4k$ steps that end there form the admissibility cycle, made of four quarters of k steps each. The residual of the initial conditions decays from the first step with the damping rate $\gamma = \ln(\alpha/|s|)$ fixed by the roots; for integer initial values it approaches zero and never reaches it, and each admissibility cycle multiplies its envelope by the same factor $e^{-4k\gamma}$, while the number four is the closure condition of the law. Harmonic Friction $F = (\lceil n/4k \rceil - n/4k) \times 100\%$ measures the position within the current cycle and is exactly zero at every boundary. The root data come from the Fibonacci Principle.

1 Recursive systems and memory depth

A system whose state advances in discrete steps of a clock, with each new state determined by the states before it, is recursive. Its memory depth k is the number of prior states required to produce the next one, and its window of k states is the foundational layer of the system: a family of depth k needs k states in place before the recurrence produces anything. Each family therefore has a layer of its own depth, two states for Fibonacci, three for Tribonacci, four for Tetranacci, and families are compared at the same multiples of their own k, not at the same raw step. The law does not state that every system is recursive; it applies to systems that advance by a clock in this way.

The results below are stated for the unweighted k-nacci families, in which every prior state enters with weight 1:

$$x(n+k) = x(n+k-1) + x(n+k-2) + \cdots + x(n). \tag{1}$$

For $k = 2$ this is Fibonacci, for $k = 3$ Tribonacci, and so on through the families listed in Table 1. The characteristic polynomial is

$$\chi_k(x) = x^k - x^{k-1} - \cdots - x - 1, \tag{2}$$

denoted $P(x)$ in the Fibonacci Principle [1]; the letter P is used here for the period. The polynomial has a dominant real root α and $k-1$ subdominant roots of smaller modulus [1, 3, 4]. The slowest subdominant root, the one of largest modulus, is denoted $s = |s|e^{\pm i\theta}$; for Fibonacci it is the real root $\psi = -1/\varphi$, with $\theta = 180°$, and from Tribonacci on it is a complex-conjugate pair [1].

Two properties of the roots are used below. The constant term of $\chi_k(x)$ is -1, so the product of all k roots has modulus 1, and no root is zero. The only rational candidates for a root are ± 1, and $\chi_k(1) = 1 - k \neq 0$ for $k \geq 2$, while $\chi_k(-1) = 1$ for even k and $\chi_k(-1) = -2$ for odd k; so α is irrational.

2 The residual and its slowest mode

From the Fibonacci Principle [1] this paper uses three things: the decomposition of the ratio error into subdominant modes (its Eq. 18), the damped-oscillator form of each mode, and the root data of its Table 1, which Table 1 here reproduces and extends to $k = 10$. For distinct roots the sequence is $x(n) = \sum_j c_j s_j^n$ with $s_1 = \alpha$. The dominant term carries the steady state of the system, and the terms with $j \geq 2$ carry what remains of the initial conditions. Since the dominant term cancels in $x(n+1) - \alpha x(n)$, the Fibonacci Principle gives the ratio error exactly in terms of these modes, and asymptotically as a sum of decaying ratios [1, Eq. 18]:

$$e_k(n) = \frac{x(n+1)}{x(n)} - \alpha = \frac{1}{x(n)} \sum_{j \geq 2} c_j (s_j - \alpha)\, s_j^n \approx \sum_{j \geq 2} A_j \left(\frac{s_j}{\alpha}\right)^n, \qquad A_j = \frac{c_j(s_j - \alpha)}{c_1}, \tag{3}$$

where the approximation uses $x(n) \approx c_1 \alpha^n$. A conjugate pair combines into one real mode $2|A|(|s|/\alpha)^n \cos(n\theta + \delta)$, which is a damped harmonic oscillator sampled at integer steps,

$$C e^{-\gamma n} \cos(\omega n + \delta), \qquad \gamma = \ln \frac{\alpha}{|s|}, \qquad \omega = \theta. \tag{4}$$

The word harmonic is used here in this sense, the damped harmonic oscillator of the Fibonacci Principle, and not in the sense of frequencies at integer multiples of a fundamental. The damping rate γ and the angle θ are fixed exactly by the roots. One full turn of the slowest mode takes

$$P = \frac{360^\circ}{\theta} \tag{5}$$

steps.

Three quantities describe the initial conditions in the present state, and they are kept distinct. The residual is the part of the state carried by the subdominant terms, $\sum_{j \geq 2} c_j s_j^n$, measured against the dominant term $c_1 \alpha^n$; it is the part of the present state that still reflects where the system started. The ratio error $e_k(n)$ is one measurement of the residual. The envelope $e^{-\gamma n}$ is the decay of the slowest subdominant mode against the dominant term. When that mode is present in the state, it sets the asymptotic size of both the residual and the ratio error, up to a constant amplitude; the envelope itself is fixed by the roots and is the same for every initial state. The initial values set only the coefficients c_j, so they affect the measured numbers and not the law. The period P and the damping rate γ are properties of the recurrence. Harmonic Friction F is a property of the cycle. The admissibility boundary $4k$ is where the two meet. None of P, γ, F and $4k$ depends on the initial values. Since no root is zero, $|s| > 0$ and the envelope is positive at every step. Since $\chi_k(1) = 1 - k < 0$ and $\chi_k(x)$ grows without bound, the dominant root satisfies $\alpha > 1$, and with $|s| < 1$ (Section 3) this gives $\gamma = \ln(\alpha/|s|) > 0$: the envelope decreases strictly from each step to the next, so its value at any step is lower than at every earlier step and higher than at every later one. Since α is irrational and every ratio $x(n+1)/x(n)$ of integer terms with $x(n) \neq 0$ is rational, the ratio error $e_k(n)$ is never exactly zero. For integer initial values not all zero, which include those of Annex A, Test 2, the dominant coefficient is never zero. If $c_1 = 0$, every remaining term has $|s_j| < 1$, so $x(n) \to 0$; an integer sequence that tends to zero is eventually zero, so the state window would eventually be the zero window, and since the companion matrix has determinant ± 1 from the constant term -1 of $\chi_k(x)$, it maps only the zero window to the zero window, which would make the initial window zero. So $c_1 \neq 0$, and the residual decays against the dominant term. The residual of the state window is also never zero: if the subdominant terms vanished from every entry of the window, each entry would equal $c_1 \alpha^m$, and the ratio $x(n+1)/x(n)$ would equal α, which

the irrationality of α rules out. Within the residual the modes superpose, and their contributions to a single entry can partly or fully cancel at a given step, as two waves of equal magnitude and opposite phase cancel at a point; the modes themselves remain, with their envelope positive and decreasing. The residual approaches zero and does not reach it; reaching zero would require the modes to stop.

The rate $|s|/\alpha = e^{-\gamma}$ is the convergence rate of power iteration on the companion matrix [5, 6], and the convergence of the ratio to the dominant root is due to Poincaré [2]. The system sheds its initial conditions continuously, from the first step, at this rate.

k	Family	θ	P	$k-P$	γ
2	Fibonacci	180.000°	2.0000	0.0000	0.962424
3	Tribonacci	124.689°	2.8872	0.1128	0.914067
4	Tetranacci	95.356°	3.7753	0.2247	0.856811
5	Pentanacci	77.038°	4.6730	0.3270	0.814033
6	Hexanacci	64.489°	5.5823	0.4177	0.783383
7	Heptanacci	55.355°	6.5035	0.4965	0.761378
8	Octanacci	48.415°	7.4357	0.5643	0.745419
9	Nonanacci	42.971°	8.3778	0.6222	0.733712
10	Decanacci	38.591°	9.3287	0.6713	0.725022

Table 1: Angle, period and damping rate of the slowest mode. The values for $k = 2$ to 8 match Table 1 of the Fibonacci Principle [1].

3 The admissibility cycle

In a window of k steps the slowest mode turns $k\theta$, which is k/P full turns.

Theorem 1 (Period of the slowest mode)**.** *For every $k \geq 2$, the slowest subdominant root of $\chi_k(x)$ has period*

$$k - 1 < P \leq k, \tag{6}$$

with equality only for Fibonacci, where $P = 2$.

The proof has three steps.

Step 1: the equation of the roots. Multiplying the characteristic polynomial by $x - 1$ gives

$$(x-1)\,\chi_k(x) = x^{k+1} - 2x^k + 1, \tag{7}$$

so every root $s \neq 1$ satisfies $s^k(2-s) = 1$. Write a subdominant root as $s = re^{i\theta}$ with $0 \leq \theta \leq 180°$; its conjugate has the same modulus and period. The equation splits into a modulus condition and an angle condition. Taking moduli, and using $|2 - s|^2 = 4 - 4r\cos\theta + r^2$,

$$g(r,\theta) = r^{2k}\left(4 - 4r\cos\theta + r^2\right) = 1. \tag{8}$$

Taking angles, $k\theta = 360°m + \varepsilon$ for an integer m, called the branch of the root, where

$$\varepsilon = \arctan\frac{r\sin\theta}{2 - r\cos\theta} \tag{9}$$

is the angle of $1/(2-s)$.

Step 2: modulus decreases as the angle increases. For $0 < r \le 1$, using $\cos\theta \le 1$,

$$\frac{\partial g}{\partial r} = r^{2k-1}\left[8k - (8k+4)\,r\cos\theta + (2k+2)\,r^2\right] \ge r^{2k-1}(2k+2)(2-r)\left(\frac{2k}{k+1} - r\right) > 0, \quad (10)$$

since $2 - r > 0$ and $2k/(k+1) > 1 \ge r$ for $k \ge 2$. So g increases with r. Also

$$\frac{\partial g}{\partial \theta} = 4r^{2k+1}\sin\theta > 0 \quad \text{for } 0 < \theta < 180^\circ, \quad (11)$$

so g increases with θ. At $\theta = 180^\circ$ this derivative is zero, and g is still strictly increasing on $(0^\circ, 180^\circ]$ because its derivative is positive on the interior. Since $g(0,\theta) = 0$ and $g(1,\theta) = 5 - 4\cos\theta > 1$ for $\theta > 0$, each angle $\theta > 0$ has exactly one modulus $r(\theta)$ in $(0,1)$ satisfying (8). The subdominant roots lie inside the unit circle [3, 4, 8], so each one is the point $r(\theta)e^{i\theta}$ at its own angle, and because g increases in both variables, $r(\theta)$ decreases strictly as θ increases over $(0^\circ, 180^\circ]$. No subdominant root is positive and real: the function $s^k(2-s)$ has derivative $s^{k-1}(2k-(k+1)s) > 0$ on $(0,1)$, so it is strictly increasing there, and it equals 1 at $s = 1$, so it is below 1 on $(0,1)$. The subdominant root of largest modulus, the slowest mode, is therefore the one of smallest positive angle.

Step 3: the branches. For any subdominant root, $r < 1$ from Step 2, and:

- Since $2 - r\cos\theta > 1$ and $r\sin\theta \ge 0$, $\varepsilon \ge 0$, with $\varepsilon = 0$ only when $\sin\theta = 0$.
- If $\theta \ge 90^\circ$, then $\varepsilon < 90^\circ \le \theta$. If $\theta < 90^\circ$, then $\tan\varepsilon < \tan\theta$ exactly when $r\cos\theta < 2 - r\cos\theta$, that is, when $r\cos\theta < 1$, which holds because $r < 1$. In both cases $\varepsilon < \theta$.

So $360^\circ m \le k\theta < 360^\circ m + \theta$, which gives

$$\frac{360^\circ m}{k} \le \theta < \frac{360^\circ m}{k-1}. \quad (12)$$

The same transformation and an intermediate-value argument give $|\theta - 360^\circ h/k| < 180^\circ/k$ for a root near each multiple $360^\circ h/k$ [7]; the branch interval places each root ahead of $360^\circ m/k$, by less than $360^\circ m/(k(k-1))$. Branch $m = 0$ would require $k\theta = \varepsilon < \theta$, which no root with $\theta > 0$ satisfies. Every root on branch $m \ge 2$ has $\theta \ge 720^\circ/k$, and $720^\circ/k \ge 360^\circ/(k-1)$ for $k \ge 2$, so every root on branch $m \ge 2$ has a larger angle than every root on branch $m = 1$. The first branch is occupied: for $k = 2$ it holds ψ, with $\theta = 180^\circ$; for $k \ge 3$, the quantity $h(\theta) = k\theta - \varepsilon(r(\theta),\theta)$ is continuous, because $r(\theta)$ is continuous on $(0^\circ, 180^\circ]$, where $\partial g/\partial r > 0$. On the closed interval $[360^\circ/k, 360^\circ/(k-1)]$, which lies in $(0^\circ, 180^\circ]$ for $k \ge 3$, the endpoint values are $h(360^\circ/k) = 360^\circ - \varepsilon \le 360^\circ$ and $h(360^\circ/(k-1)) > 360^\circ k/(k-1) - 360^\circ/(k-1) = 360^\circ$, so $h = 360^\circ$ at some θ in the interval, and not at the right endpoint, where $h > 360^\circ$; that θ lies in $[360^\circ/k, 360^\circ/(k-1))$, and the point $r(\theta)e^{i\theta}$ satisfies both conditions of Step 1. By Step 2, the slowest mode lies on the first branch, and on that branch

$$360^\circ \le k\theta < 360^\circ + \theta, \quad (13)$$

which gives $\theta \ge 360^\circ/k$, so $P \le k$, and $\theta < 360^\circ/(k-1)$, so $P > k - 1$. Equality $P = k$ requires $\varepsilon = 0$, hence $\theta = 180^\circ$, and a first-branch angle of 180° requires $360^\circ/k \le 180^\circ < 360^\circ/(k-1)$, which holds only for $k = 2$. □

For Tribonacci, $r = 0.7373527$ and $\theta = 124.689^\circ$, so $\sin\theta = 0.82230$, $\cos\theta = -0.56902$, and $\varepsilon = \arctan(0.60633/2.41957) = \arctan 0.25059 = 14.067^\circ$; then $360^\circ + 14.067^\circ = 374.067^\circ = 3 \times 124.689^\circ$, and θ lies in $[120^\circ, 180^\circ)$, the first branch for $k = 3$.

So one window of k steps holds one full turn of the slowest mode for Fibonacci and slightly more than one turn for $k \geq 3$, and the shortfall $k - P$ grows toward 1 as k increases (0.1128 at $k = 3$, 0.6713 at $k = 10$, 0.9959 at $k = 100$ and 0.9998 at $k = 400$).

The admissibility cycle has length $4k$ and consists of four quarters of k steps each. Two angles are involved and they are distinct quantities. Each quarter is 90° of the cycle, so the four quarters complete $4 \times 90° = 360°$ of the cycle at $n = 4k$. Within each quarter the slowest mode completes k/P turns of its own phase: exactly one for Fibonacci, and for every other family more than one and less than $k/(k-1)$, at most 1.0763 among the families computed ($k = 7$). By $n = 4k$ the mode has completed $4k/P$ turns, so four full turns are complete by the end of the fourth quarter. The quarters belong to the cycle, the turns belong to the mode, and the boundary $4k$ is where the two are compared.

The admissibility boundary. Corollary 1. *For every $k \geq 2$, $3k < 4P \leq 4k$, so $n = 4k$ is the first window boundary at which the slowest mode has completed four full turns.* The fourth full turn of the slowest mode completes at $n = 4P$. From $P \leq k$ in (6),

$$4P \leq 4k. \tag{14}$$

From $P > k - 1$, $4P > 4k - 4$, and $4k - 4 \geq 3k$ whenever $k \geq 4$. For $k = 2$, $4P = 8 > 6$, and for $k = 3$, the computed value $4P = 11.549$ exceeds 9. Therefore

$$3k < 4P \leq 4k \tag{15}$$

for every $k \geq 2$. The fourth turn is not complete at the third window boundary $n = 3k$ and is complete by the fourth, $n = 4k$. The step $n = 4k$ is the first window boundary at which the slowest mode has completed four full turns. This step is the admissibility boundary.

For Tribonacci, $P = 360°/124.689° = 2.8872$, so $4P = 11.549$, which lies between $3k = 9$ and $4k = 12$; at $n = 12$ the mode has turned $12 \times 124.689° = 1496.27°$, which is $1496.27/360 = 4.156$ turns. For Fibonacci the count is exact: $8 \times 180° = 1440° = 4 \times 360°$.

k	$4P$	$\lceil 4P \rceil$	$4k$	Turns at $4k$	$e^{-4k\gamma}$
2	8.000	8	8	4.000	4.53×10^{-4}
3	11.549	12	12	4.156	1.72×10^{-5}
4	15.101	16	16	4.238	1.11×10^{-6}
5	18.692	19	20	4.280	8.50×10^{-8}
6	22.329	23	24	4.299	6.84×10^{-9}
7	26.014	27	28	4.305	5.51×10^{-10}
8	29.743	30	32	4.304	4.37×10^{-11}
9	33.511	34	36	4.297	3.38×10^{-12}
10	37.315	38	40	4.288	2.54×10^{-13}

Table 2: Completion of the fourth turn ($4P$), the first step at which it is complete ($\lceil 4P \rceil$), the admissibility boundary ($4k$), the number of turns at the boundary ($4k/P$), and the envelope of the residual at the boundary.

The boundary is a statement about windows. The first single step at which four turns are complete is $\lceil 4P \rceil$, which equals $4k$ for $k = 2, 3, 4$ and falls before $4k$ from $k = 5$ on, since four turns are complete at $4k - 1$ exactly when $4P \leq 4k - 1$, that is when $k - P \geq 1/4$, and $k - P = 0.3270$ at $k = 5$. The step $4k$ is the first multiple of k after four full turns in every family.

A window of $4k$ steps holds $4k/P$ turns of the slowest mode: exactly four for Fibonacci, and between 4.010 and 4.305 for the other families computed, an excess of $4(k-P)/P$ of a turn.

The admissibility cycle is closed by construction: a cycle is what returns to itself, and its closure is a tautology in the way a circle is. The law states where the k-nacci roots complete that return. The period P and the damping rate γ come from the roots of $\chi_k(x)$, the four quarters come from the cycle, and the boundary $4k$ is where the two meet.

Remark. Gómez and Luca conjecture that every root's argument lies within $1/k$ radian of a multiple of $2\pi/k$ [8]. The branch interval gives $0 \le \theta - 2\pi m/k < 2\pi m/(k(k-1))$, which is below $1/k$ whenever $2\pi m < k-1$, so the conjectured bound holds for every root on such a branch, and for the slowest root at every $k \ge 8$. For $k = 3$ to 7 the slowest root's gaps are 0.0818, 0.0935, 0.0879, 0.0783 and 0.0685 radian, each below $1/k$.

4 The residual at the boundary

At the admissibility boundary the envelope of the residual is $e^{-4k\gamma}$ (Table 2). Each admissibility cycle multiplies the envelope by this same factor. After j cycles,

$$e^{-4jk\gamma} = \left(e^{-4k\gamma}\right)^j, \tag{16}$$

and after half a cycle, $e^{-2k\gamma} = (e^{-4k\gamma})^{1/2}$. For Fibonacci, $\gamma = 0.962424$, so one cycle gives $e^{-8\gamma} = e^{-7.699392} = 4.53\times10^{-4}$ and two cycles give $(4.53\times10^{-4})^2 = 2.05\times10^{-7} = e^{-16\gamma}$. The multiplicative relation is the general property $e^{-(m+n)\gamma} = e^{-m\gamma}e^{-n\gamma}$ of exponential decay; what the law fixes is the unit, the cycle of four quarters ending at $4k$. The residual does not reset at a boundary. The factor repeats: each cycle acts on what the previous cycle left, and composing cycles multiplies their factors.

When the slowest mode is present in the state, the ratio error follows the envelope up to a constant amplitude and oscillates with a phase set by the roots and the initial values. Table 3 measures the actual ratio error $|e_k(n)|$ with each family starting from its own foundational layer: $x(0) = \cdots = x(k-2) = 0$ and $x(k-1) = 1$, so that the first computed value, $x(k)$, is the first step at which a full window of k states exists. This is the convention of the Fibonacci Principle [1].

k	$\lvert e(3k)\rvert$	$\lvert e(4k)\rvert$	$\lvert e(4k+1)\rvert$	Smallest in $(3k, 4k]$	$\max_{4k<n<12k} \lvert e(n)\rvert/\lvert e(4k)\rvert$
2	6.97×10^{-3}	1.01×10^{-3}	3.87×10^{-4}	$n = 8$	0.382
3	1.62×10^{-3}	1.29×10^{-4}	1.04×10^{-6}	$n = 12$	0.157
4	3.23×10^{-4}	1.46×10^{-5}	3.05×10^{-6}	$n = 14$	0.209
5	6.05×10^{-5}	1.58×10^{-6}	5.63×10^{-7}	$n = 20$	0.355
6	1.09×10^{-5}	1.65×10^{-7}	7.61×10^{-8}	$n = 24$	0.461
7	1.90×10^{-6}	1.65×10^{-8}	8.91×10^{-9}	$n = 28$	0.540
8	3.19×10^{-7}	1.58×10^{-9}	9.49×10^{-10}	$n = 32$	0.600

Table 3: Measured ratio error around the admissibility boundary, each family starting from its own foundational layer.

The statement that holds for every family and every choice of initial values is the envelope statement: at $n = 4k$ the envelope is $e^{-4k\gamma}$, lower than at every earlier step, and it decreases by the factor $e^{-4k\gamma}$ from each boundary to the next. Table 3 illustrates it with measured values for

one choice of initial values. In every family the measurement at $4k$ is smaller than at $3k$, by a factor between 6.9 and 202, and every measurement from $4k+1$ to $12k$ lies below the measurement at $4k$. For Fibonacci the value at the boundary is exact: from the Fibonacci Principle, $e(n) = (-1)^n\sqrt{5}/(\varphi^{2n} - (-1)^n)$ [1], and at $n = 8$ this is $2.2360680/(2206.99955 - 1) = 1.01363 \times 10^{-3}$, which equals $34/21 - \varphi$. Where the smallest measurement falls inside the closing quarter is set by the phase of the mode, and the phase depends on the initial values. It falls at $4k$ in six of the seven families; in Tetranacci it falls at $n = 14$, with $|e(14)| = 6.99 \times 10^{-6}$ against $|e(16)| = 1.46 \times 10^{-5}$. If instead every family is given one shared starting index, $x(1) = 1$ with zeros before it, the measured values change: the dip moves to Tribonacci, where $|e(12)| = 1.04 \times 10^{-6}$ and steps 13 to 16 measure up to 19.5 times above it, while Tetranacci reaches its smallest value at $n = 16$. The measured values follow the initial values.

The ratio $x(n+1)/x(n)$ converges to α, which is an equilibrium in the exact sense, the fixed point of the ratio map, and the shape of the state window settles onto the dominant mode, which is the steady state of the system. After the boundary the system continues to oscillate around α with an amplitude that decreases at rate γ.

5 Harmonic Friction

Harmonic Friction F measures the position of the system within the current admissibility cycle, as the fraction of the cycle that remains before the next boundary:

$$F(n) = \left(\left\lceil \frac{n}{4k} \right\rceil - \frac{n}{4k} \right) \times 100\%. \tag{17}$$

In the first cycle, $0 < n \leq 4k$, the ceiling equals 1 and the definition reduces to

$$F(n) = \left(1 - \frac{n}{4k} \right) \times 100\% = \left| \frac{n}{4k} - 1 \right| \times 100\%, \tag{18}$$

which is the original form of the measure. The ceiling extends it to every cycle, so that $F = 0$ at each boundary $n = 4k, 8k, 12k, \ldots$ and F begins again after each one. With $k = 4$ and $4k = 16$: at $n = 4$, $F = 1 - 0.25 = 75\%$; at $n = 8$, $F = 50\%$; at $n = 12$, $F = 25\%$; at $n = 15$, $F = 1/16 = 6.25\%$; at $n = 16$, $F = \lceil 1 \rceil - 1 = 0\%$; and at $n = 17$, $F = \lceil 1.0625 \rceil - 1.0625 = 93.75\%$. The completed fraction of the cycle is $100\% - F$, which is 100% at the boundary.

F and the residual are distinct measurements. F is exactly zero at every boundary and repeats with each cycle; the residual is near zero and nonzero at every boundary, and its envelope decreases by the factor $e^{-4k\gamma}$ from one boundary to the next. F locates the system in its cycle, and the residual measures how much of the initial conditions remains. F is a diagnostic scalar read from the state at a given step and is not a term of the recurrence.

6 The Law of Admissibility

Law of Admissibility. For a recursive system of memory depth k in the k-nacci family, the admissibility cycle has length $4k$ and consists of four quarters of k steps. The slowest damped mode has period P with $k-1 < P \leq k$, so its fourth full turn completes at $4P$ with $3k < 4P \leq 4k$, and $n = 4k$ is the first window boundary after four full turns: the admissibility boundary. At every boundary $n = 4jk$, Harmonic Friction is zero and the envelope of the residual of the initial conditions is $(e^{-4k\gamma})^j$, near zero and nonzero.

k	Family	$4k$	$F(4k-1)$	$F(4k)$	$F(4k+1)$
2	Fibonacci	8	12.50%	0%	87.50%
3	Tribonacci	12	8.33%	0%	91.67%
4	Tetranacci	16	6.25%	0%	93.75%
5	Pentanacci	20	5.00%	0%	95.00%
6	Hexanacci	24	4.17%	0%	95.83%
7	Heptanacci	28	3.57%	0%	96.43%
8	Octanacci	32	3.13%	0%	96.88%
9	Nonanacci	36	2.78%	0%	97.22%
10	Decanacci	40	2.50%	0%	97.50%

Table 4: Harmonic Friction around the admissibility boundary: one step before, at, and one step after it. $F(4k-1) = 1/4k$ and $F(4k+1) = 1 - 1/4k$.

The constant $R = 4$ is the number of quarters in the cycle and the number of full turns of the slowest mode completed at the boundary. At $n = 4k$ the mode has turned $4k/P$ times, and from $k - 1 < P \leq k$, $4 \leq 4k/P < 4k/(k-1)$, which is at most 5 for $k \geq 5$; for $k = 2, 3, 4$ the values are 4.000, 4.156 and 4.238. So exactly four full turns are complete at the boundary in every family, with a partial fifth turn under way for $k \geq 3$. The roots fix the period P and the damping rate γ, and with them the step at which four full turns are complete; the number four is the closure condition of the law.

7 Conclusion

The admissibility boundary $n = 4k$ is the first window boundary at which the slowest damped mode of a k-nacci system has completed four full turns, because the period of that mode satisfies $k - 1 < P \leq k$, for every $k \geq 2$, and therefore $3k < 4P \leq 4k$. The residual of the initial conditions decays from the first step at the damping rate $\gamma = \ln(\alpha/|s|)$; at each boundary its envelope is $e^{-4k\gamma}$ times its value at the previous boundary, while the measured values oscillate beneath that decay with a phase set by the initial values. Harmonic Friction $F = (\lceil n/4k \rceil - n/4k) \times 100\%$ locates the system within its cycle and is exactly zero at every boundary. The quantities that place the boundary come from the roots, and the law states where those roots complete the four-quarter cycle. The roots, their location inside the unit circle and the rate of convergence are classical [2–8]; the period bound of the slowest mode, the admissibility boundary and the exact count of four full turns are derived in this paper.

Annex A: Test of the stated results

The derivations in this paper are checked, and the measurements obtained, by three computations. Each is described by its inputs, its operations, the criterion it checks and its result.

Test 1: roots and periods. Inputs: the polynomial $\chi_k(x) = x^k - x^{k-1} - \cdots - 1$ for every k from 2 to 400. Operations: compute all roots; take the real root of largest modulus as α and the remaining root of largest modulus as s; compute $\theta = |\arg s|$, $P = 360°/\theta$ and $\gamma = \ln(\alpha/|s|)$. Criteria: every subdominant root satisfies the modulus condition (8) and the branch interval $360°m/k \leq \theta <$

$360°m/(k-1)$; modulus decreases as angle increases; the slowest root lies on the first branch; $k-1 < P \leq k$; and $k\theta - 360° = \varepsilon$, checked to 50 significant digits at $k = 3, 4, 5, 6, 8, 10, 20, 50, 100$ and 400. Result: every criterion holds at every k checked (the branch and ordering criteria for every subdominant root from $k = 2$ to 200, the period criteria from $k = 2$ to 400), with $P = k$ only at $k = 2$, in agreement with the derivation of Section 3. Tables 1 and 2 list $k = 2$ to 10.

Test 2: ratio error from each family's foundational layer. Inputs: the unweighted recurrence for $k = 2$ to 8 with $x(0) = \cdots = x(k-2) = 0$ and $x(k-1) = 1$, so that each family begins with its own window of k states, generated in exact integer arithmetic through $n = 12k$; α computed to 80 significant digits. This is the convention of the Fibonacci Principle [1]. The same computation was repeated with one shared starting index for all families, $x(1) = 1$ and $x(n) = 0$ for $2-k \leq n \leq 0$, and its differences are reported in Section 4. Other initial values or another indexing give other measured values, while P, γ and the boundary are unchanged (Section 2). Operations: compute $|e_k(n)| = |x(n+1)/x(n) - \alpha|$ at every step. Criteria: the value at $4k$ compared with the value at $3k$; the step of smallest value in $(3k, 4k]$; every value for $4k < n < 12k$ compared with the value at $4k$. Result: Table 3.

Test 3: correlation with random initial conditions. Inputs: the unweighted recurrence for $k = 3$ to 8, with the k initial values drawn independently from a standard normal distribution, 20,000 trials per family. Operations: at each step n, compute the Pearson correlation between the current window of k states and the initial window, and average over trials; take the value at $12k$ as the reference floor. Criterion: the distance from the floor at $2k$, $3k$, $4k$, $6k$ and $8k$. Fibonacci is not included, because the correlation between two windows of length 2 is always $+1$ or -1 and cannot resolve the approach to the floor; this is a limit of the correlation measure, and the roots place the boundary for Fibonacci as for every other family.

k	Floor ($12k$)	$2k$	$3k$	$4k$	$6k$	$8k$
3	0.1104	2.5×10^{-1}	2.3×10^{-2}	1.6×10^{-3}	2.0×10^{-6}	3.4×10^{-8}
4	0.0873	1.5×10^{-1}	6.3×10^{-3}	1.4×10^{-4}	1.4×10^{-7}	3.8×10^{-10}
5	0.0567	8.8×10^{-2}	1.7×10^{-3}	1.6×10^{-5}	5.4×10^{-9}	2.7×10^{-12}
6	0.0517	5.1×10^{-2}	5.1×10^{-4}	1.4×10^{-6}	3.0×10^{-10}	3.1×10^{-14}
7	0.0367	2.8×10^{-2}	1.6×10^{-4}	2.7×10^{-7}	9.3×10^{-12}	2.7×10^{-16}
8	0.0276	1.6×10^{-2}	4.8×10^{-5}	3.4×10^{-8}	4.9×10^{-13}	$< 10^{-16}$

Table 5: Distance of the averaged window correlation from its reference floor at $12k$.

Result: over the tested interval the averaged correlation settles toward a nonzero reference value measured at $12k$, as the window in each trial settles onto the shape of the dominant mode, and the distance from the reference floor at $3k$ is between 14 and 1,400 times the distance at $4k$. At $8k$, the second boundary, every family is within 3.4×10^{-8} of its floor, and from $k = 7$ on the distance reaches the resolution of 64-bit floating-point arithmetic. Values at that resolution depend on the precision of the arithmetic used, while P, γ and the boundary come from the roots and do not.

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